

RYAN P. SEAMAN

ryan_seaman@hms.harvard.edu

Boston, MA | (503) 781-7821 | <https://rpseaman.github.io>

EDUCATION

Harvard Medical School
MMSc. Biomedical Informatics

Boston, MA
Expected Graduation May 2027

Relevant Coursework:

Foundations of Biomedical Informatics (BMI701), Data Visualizations for Biomedical Applications (BMI706), Advanced Coding and Statistics for Biomedical Informatics (BMI714), Conduct and Communication of Science (BMI717), Data Science (AC209a)

Colby College
B.A. Computational Biology
Varsity Soccer, 4 years

Waterville, ME
January 2023

*Relevant Coursework: *AP credit ** UT Austin/EdX*

Biology/Chemistry: General Chemistry, Organic Chemistry, General Biology*, Comparative Biomechanics, Animal Behavior, Evolutionary Analysis, Genetics, Genomics, Applied Biomedical Genomics, Morphometry (R), Systems Biology (Python)

Math/Statistics: Calculus*, Statistics*, Linear Algebra**, Intro to Statistics and Data Science (R)

Computer Science: Computational Thinking for Science (Python), Data Structures and Algorithms (Java), Data Analysis and Visualization (Python), Computer Organization (VHDL), Software Engineering (many), Artificial Intelligence Seminar (Python), Computing Ethics

EXPERIENCE

Bioinformatician, Comparative Genomic and Data Science Core
MDI Biological Laboratory

January 2023 – August 2025
Bar Harbor, ME

Collaborate with research labs to analyze data, providing customized bioinformatics solutions to enhance project outcomes

- Developed and implemented data analysis tools that empower researchers to explore and analyze their own data, improving research efficiency
- Created an open-source single-cell RNA sequencing pipeline for downstream analysis, ensuring rigorous, reproducible data analysis
- Designed and maintained cloud infrastructure for analysis and secure data storage
- Tested and integrated new microscopy analysis techniques for the MDIBL Microscopy Core to enhance research capabilities
- Led educational programs, including internal trainings and courses for the broader research community on bioinformatics tools and techniques

Research Assistant, Chang Lab
Department of Geology, Portland State University

May – August 2017
Portland, OR

- Analyzed water quality trends along the Clackamas River using statistical methods to investigate climate change impacts on water resources
- Presented findings to the department to contribute to ongoing regional climate change studies

PEER REVIEWED PUBLICATIONS

Seaman RP, Campbell R, Doe V, Yosufzai Z, Graber JH. A cloud-based training module for efficient de novo transcriptome assembly using Nextflow and Google Cloud. *Brief Bioinform.* 2024;25(4):bbbe313. doi:10.1093/bib/bbae313.

PREPRINT PUBLICATIONS

García-García, D, Knapp, D, Kim, M, Jamwal, K, Fuqua, H, **Seaman, RP**, Grindle, RE, Nowoshilow, S, Novatchkova, M, Kolling, FW, Graber, JH, Murawala, P. The essential role of connective-tissue cells during axolotl limb regeneration. *bioRxiv*, 2025.03.30.645595. doi:10.1101/2025.03.30.645595.

Paulmann A, Cox MD, Boewer T, Somers HM, Fuqua H, **Seaman RP**, Graber JH, Mahajan A, Johnson CP, Beverly-Staggs LL, Sandhi S, Schenk H, Haller H. SGLT2 Inhibition Ameliorates Age-Dependent Renovascular Rarefaction. *bioRxiv [Preprint]*. 2025 Jun 28:2025.06.27.654312. doi: 10.1101/2025.06.27.654312.

MANUSCRIPTS IN PROGRESS

“Macrophage trajectories in axolotl limb regeneration” — *Godwin Lab, MDIBL*

“Renal regeneration in zebrafish: sources of regenerative cells and mechanisms of differentiation and dedifferentiation” — *Drummond Lab, MDIBL*

“Early-life exposure to cortisol: implications in zebrafish development and regeneration” — *Coffman Lab, MDIBL*

WORKS IN PROGRESS

“Cellular transitions and tissue remodeling in axolotl skin during forced metamorphosis and limb regeneration” — *Murawala Lab, MDIBL* (data collected, analysis in progress)

“Differential kidney function in neotenic and metamorphic axolotl” — *Murawala Lab, MDIBL and Hinze Lab, Hanover Medical School* (data collected analysis in progress)

“Divergent mechanisms of muscular regeneration in axolotl limb and tail regeneration” — *Murawala Lab, MDIBL* (data collected analysis in progress)

“SCSACPE: scRNAseq pipeline for downstream analysis” — *CGDS Core, MDIBL* (pipeline developed)

TEACHING EXPERIENCE

Course Faculty — ScRNAseq in the Cloud: Pipelines to Insights, October 2024
MDI Biological Laboratory

- Co-taught an intensive 3-day course on using cloud infrastructure to process single-cell RNA sequencing data for bioinformaticians and biologists
- Guided participants through workflows from raw sequence files to dimensionally reduced projections
- Emphasized best practices for rigorous, reproducible analysis in bioinformatics workflows
- Created instructional content and built course website: mdibl.github.io/scrnaseq_course_2024/

Instructor — ReBILD Course: Emerging Leaders in Regenerative Biology and Development, August 2024/2025

MDI Biological Laboratory

- Led a hands-on session on single-cell RNA sequencing data analysis, training participants in the use of the Loupe Browser tool for exploring and interpreting results

Summer Intern Instructor and Mentor, Summers 2023, 2024, and 2025

MDI Biological Laboratory

- Taught introductory classes in Python programming for biological applications to all summer interns
- Held office hours to provide personal guidance on bioinformatics tools and methodologies
- Mentored bioinformatics interns on their computational projects

ADDITIONAL EXPERIENCE

Colby College — Admissions Assistant Coordinator, Summer 2021

Gave campus tours to prospective students and families. Promoted into leadership role where I coordinated all tours for the summer tour guide cohort and assisted with other admissions office tasks

Lake Oswego Soccer Club — Youth Team Coach, Summers 2018-2020

Developed and mentored players for the local soccer club

SKILLS and TECHNIQUES

Analysis modalities: RNAseq, scRNAseq, 10x Visium, 10x Xenium, 10x Multiome (scRNAseq + scATACseq)

Programming languages: Python, R, Java, Groovy, Bash

Bioinformatics tools: Nextflow, nf-core, command line, genomics analysis tools

Cloud computing platforms: Google Cloud Platform, Amazon Web Services

Data analysis and visualization: R, Python, Excel

Other skills: teaching/mentoring, computing ethics, Git/GitHub

LANGUAGES

Mandarin Chinese (conversational)